THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE CHAPTER TEST SET B

Chapter 1: **Chemical Reactions and Equations** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. Which of the following is a physical change? 1

(a) melting of ice **(b)** burning of paper
(c) digestion of food **(d)** rusting of iron

2. $2\,H_2 + O_2 \longrightarrow 2\,H_2O$ is a: 1

(a) displacement reaction **(b)** double displacement reaction
(c) decomposition reaction **(d)** combination reaction

3. The burning of natural gas, $CH_4 + 2\,O_2 \longrightarrow CO_2 + 2\,H_2O$, is: 1

(a) a displacement **(b)** a decomposition **(c)** endothermic **(d)** exothermic

4. A student states: “In a balanced equation the number of molecules on both sides must be equal.” The correct response is: 1

(a) Correct
(b) Correct only for gases
(c) Incorrect — the number of atoms of each element must be equal
(d) Cannot be determined

5. For $CaCO_3 \longrightarrow CaO + CO_2$, Student A says decomposition; Student B says combination. Then: 1

(a) Student A is correct **(b)** Student B is correct
(c) Both are correct **(d)** Both are wrong

6. The brown gas evolved on heating lead nitrate is: 1

(a) NO **(b)** O_2 **(c)** NO_2 **(d)** N_2

7. When silver nitrate solution is added to sodium chloride solution, the white precipitate formed is: 1

(a) silver oxide **(b)** silver chloride **(c)** sodium nitrate **(d)** sodium chloride

8. Oxidation is: 1

(a) gain of electrons **(b)** gain of hydrogen
(c) loss of oxygen **(d)** gain of oxygen or loss of hydrogen

9. The blue colour of copper sulphate solution fades when an iron nail is dipped in it because: 1

(a) iron displaces copper, forming $FeSO_4$ **(b)** copper displaces iron
(c) water evaporates **(d)** the nail dissolves completely

10. When copper powder is heated in air, its surface turns black due to the formation of: 1

(a) $CuSO_4$ **(b)** $CuCO_3$ **(c)** CuO **(d)** Cu_2S

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Why is respiration considered an exothermic reaction? 2

12. Balance the thermite reaction and name its type: $Fe_2O_3 + Al \longrightarrow Al_2O_3 + Fe$. 2

13. Why are decomposition reactions called the opposite of combination reactions? Give an example. 2

14. A student says silver chloride turns grey in sunlight due to oxidation of silver. Justify or refute. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 30–40 words each.

15. Write balanced equations with state symbols: 3
(i) barium chloride and sodium sulphate solutions react
(ii) sodium hydroxide reacts with hydrochloric acid
(iii) zinc reacts with silver nitrate solution.

16. Identify the substance oxidised and reduced: 3
(i) $4\,Na + O_2 \longrightarrow 2\,Na_2O$
(ii) $CuO + H_2 \longrightarrow Cu + H_2O$
(iii) $MnO_2 + 4\,HCl \longrightarrow MnCl_2 + 2\,H_2O + Cl_2$

17. Differentiate between exothermic and endothermic reactions with one example each. 3

OR

Calcium oxide reacts vigorously with water. 3
(i) Name the product.
(ii) Name the type of reaction.
(iii) Why do walls whitewashed with it become shiny after a few days?

18. In the electrolysis of water: 3
(i) Name the gases collected at the cathode and anode.
(ii) What is the ratio of their volumes?
(iii) Why are a few drops of dilute sulphuric acid added?

OR

Why do silver articles turn black and copper articles turn green on exposure to air? Name the substances formed. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. Identify the type of each reaction: 5
(a) $2\,KClO_3 \longrightarrow 2\,KCl + 3\,O_2$
(b) $Fe + CuSO_4 \longrightarrow FeSO_4 + Cu$
(c) $Na_2SO_4 + BaCl_2 \longrightarrow BaSO_4 + 2\,NaCl$
(d) $2\,Mg + O_2 \longrightarrow 2\,MgO$
(e) $ZnO + C \longrightarrow Zn + CO$

OR

(i) Define corrosion. 5
(ii) Describe an activity with three test tubes to show that both air and water are needed for rusting.
(iii) Give one method to prevent rusting.

20. **Case study:** A shiny brown element X turns into a black compound Y when heated in air. 5
(i) Identify X and Y.
(ii) Write the equation.
(iii) Name the type of reaction.
(iv) How can Y be converted back to X? Write the equation.

OR

A white salt on heating gives brown fumes and leaves a yellow residue. 5
(i) Name the salt.
(ii) Write the equation.
(iii) Name the type of reaction.
(iv) Name the brown gas.
(v) Name the yellow residue.

END OF QUESTION PAPER Total: 40 Marks

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